

Agency and Creativity in GenAI-based Game Development and Design Education

Gaole Huang

Beijing Mayfly & Grain Technology Co., Ltd.

Beijing, China

huangcola@hotmail.com

Jian Tan

School of Digital Media & Design Arts

Beijing University of Posts and Telecommunications (BUPT)

Beijing, China

tanjian@bupt.edu.cn

Max Chen

Worcester Polytechnic Institute

Worcester, Massachusetts, United States

mjchen@wpi.edu

Abstract—The rapid development of generative artificial intelligence (GenAI) is reshaping the game design and development industry, creating an urgent need to prepare game developers who can effectively apply GenAI in creative practice. However, research on GenAI game development education remains limited, particularly regarding how GenAI influences game developers' creative agency during training. This paper presents a qualitative study on 30 projects and reflection reports from a graduate course, Artificial Intelligence-Generated Content (AIGC) Game Development. We examine how GenAI shapes students' creative agency in AI-assisted game development, drawing on theories of intrinsic motivation. Based on our findings, we propose a framework for designing GenAI game development education to help students become creators who can actively and critically engage with GenAI.

Index Terms—Generative AI (GenAI), Artificial Intelligence Generated Content (AIGC), Game Development Education, Creative Agency, Human-AI Collaboration, Intrinsic Motivation

I. INTRODUCTION

Generative Artificial Intelligence (GenAI) is reshaping game development by enabling rapid prototyping, asset generation, and new forms of human-AI collaboration, offering novice developers a way to bridge gaps between creative intent and technical execution. This paper presents an exploratory case study of a graduate-level Artificial Intelligence-Generated Content (AIGC) Game Development course, using thematic analysis of 30 student projects and reflections to examine how GenAI shapes students' agency and creativity. We find that GenAI supports prototyping and skill extension but can weaken agency when students surrender authorship to the tool or compromise the original design vision. Based on these findings, we propose a heuristic framework to help educators identify when GenAI supports or constrains students' autonomy, competence, and relatedness during game creation. Our

work contributes to ongoing conversations about game design and development education in the GenAI era. We argue that game education should not frame GenAI as something students must simply adopt or reject, but should instead help them become intentional, reflective, and self-directed creators who can work critically with GenAI.

II. RELATED WORK

Our work builds on research in AI-assisted game creation, GenAI-supported development practice, and game development education.

A. AI-Assisted and Mixed-Initiative Game Creation

Procedural content generation (PCG) and mixed-initiative systems have long established that designers retain final responsibility for evaluating and integrating AI-generated content into coherent play experiences [1], [2]. While explainable AI research further emphasizes the need for controllable, domain-specific design tools [3], most contemporary GenAI tools are general-purpose and black-boxed, posing particular barriers for novice learners [4].

B. Generative AI in Game Development Practice

Industry-focused research finds that developers view GenAI as a collaborative tool while acknowledging its limitations, including inconsistent generation and lack of the “soul” that only human creators can bring [5], [6]. Early-career game developers' responses to GenAI in a professional development program show that adoption is shaped not only by usability and workflow fit, but also by ethical skepticism, concerns about agency and ownership, and emerging professional identities [7]. GenAI adoption is highly contextual, as perceptions shift depending on factors such as experience, cultural context, the proportion of artists in the community, and broader societal normalization of GenAI [8]. These studies highlight practical and ethical concerns around workflow integration, trust, authorship, collaboration, and professional identity in small-scale

Correspondence concerning this paper can be addressed to Max Chen at mjchen@wpi.edu.

game development. However, less is known about how novice developers *learn* to work with GenAI in formal educational settings.

III. RESEARCH METHOD

We conducted an exploratory case study to examine students' engagement with GenAI in a graduate-level game development course. This allowed us to document situated practices, identify emerging patterns, and develop interpretive insights about AI-assisted learning rather than test predetermined hypotheses [9].

A. Research Context and Data Collection

The study was conducted in a 16-week graduate-level "Artificial Intelligence Generated Content (AIGC) Game Creation" class at Beijing University of Posts and Telecommunications (BUPT) in Fall 2025. 30 students enrolled in this class from diverse disciplinary backgrounds, including 16 with little or no prior programming experience. Students developed final game projects using GenAI tools of their choice, including Google AI Studio/Gemini and DeepSeek for programming support, and Nano Banana Pro, Midjourney, and DALL-E 3 for visual asset generation.

The dataset included de-identified course materials from final projects, such as playable games, design documents, assignments, and presentation or defense recordings. The study was reviewed and deemed exempt by the host university, and informed consent was obtained from all participating students.

B. Qualitative Analysis

We conducted thematic analysis following Braun and Clarke's six-phase method [10]. Two researchers independently generated initial codes capturing patterns of GenAI use, resistance, and evaluation, then iteratively refined and grouped them into themes through team discussion and dataset validation. All authors approved the final thematic structure.

IV. FINDINGS

We identified three themes that characterize how students interacted with GenAI during game development. These themes describe different patterns of use, non-use, evaluation, adaptation, and integration. Note that these patterns are not mutually exclusive; a student's project could exhibit multiple behaviors.

A. Use and Non-Use of AI Assistance

Students made different choices about when to use GenAI, what tasks to assign to it, and when to avoid it. These choices were shaped by project goals, prior experience, perceived tool reliability, and students' views about the role of human authorship in game creation.

1) Using AI for Generation and Implementation Support:

All students used GenAI to generate implementation plans, code, visual assets, dialogue, or real-time game content. However, 8 out of 30 students failed to produce playable demos and instead submitted work-in-progress or PowerPoint presentations for their final reports. Project completeness often depends on how students evaluate, modify, and integrate the generated content into their project design.

2) *Choosing Not to Use AI for Certain Tasks:* 6 out of 22 students who submitted their demos chose not to use GenAI for particular aspects of their projects, especially core narrative, artistic, or conceptual elements. 2 of them explained that "the core of the story must be thought and conveyed by a human," and "AI art lacks the human touch." Students used AI selectively, setting clear boundaries around what they wanted to create themselves and what they were willing to delegate.

B. GenAI as Design Driver

Students allowed AI-generated outputs to strongly shape the direction, scope, or implementation of the final game. These patterns were observed when students accepted outputs without substantial modification or compromised their design vision to fit what the tool could produce.

1) *Accepting AI Outputs with Limited Revision:* 12 out of 30 students incorporated AI-generated content with limited verification or revision. For example, these students either used tools to generate entire games from a single prompt, or accepted mechanics or algorithms for pricing systems or rhythm detection. In these cases, the final projects often contained elements that students could not explain in their own words. This pattern also occurred when asset production preceded the full consolidation of game design. Fig. 1 shows an example of a visually complex map that did not account for level design because that GenAI-generated asset was incorporated before its gameplay function had been clearly specified.



Fig. 1. A student generated visually complex level maps using GenAI. However, the submitted project did not clearly specify the underlying mechanics, traps, or puzzle logic needed to make the level playable.

2) *Compromising Game Design Due to GenAI:* 6 out of 30 students changed project goals when GenAI could not produce the desired output. For example, one student initially designed a skill-based "Cake Baking Simulator" but shifted to a simpler decoration game because GenAI could not deliver the required physics code. Another student used an AI-generated video that differed from the intended narrative sequence, in which a horse appeared to fall behind rather than lead. These cases show how tool limitations influenced the scope and direction of student projects. In addition, the inconsistency in GenAI-produced outcomes may lead to an unexpected gap between design and implementation, as shown in Fig. 2.

C. GenAI Outputs as Iterative Materials

Students treated AI outputs as intermediate materials that required testing, revision, and integration. These students used GenAI to support production, but continued to make design decisions about what to keep, modify, discard, or verify.

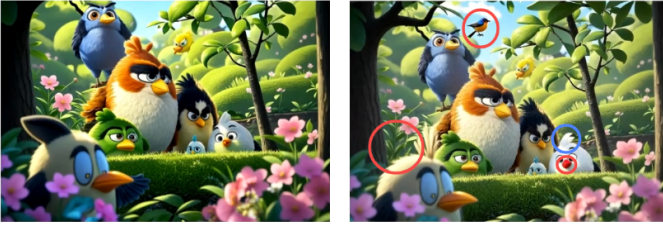


Fig. 2. In a “spot the difference” game, the intended design included three differences marked by red circles. The AI-generated image introduced an additional unintended difference, marked by the blue circle, that was not accounted for in the game’s target-checking logic.

1) *Evaluating and Revising AI Outputs*: 14 out of 30 students reviewed AI-generated outputs against their design goals and adjusted them when the outputs did not fully align with the intended experience. This included using AI-generated art as placeholders, manually adjusting generated assets, checking generated code, or asking peers to review AI-assisted implementation. These practices show how students combined AI assistance with human evaluation and revision. Fig. 3 shows an example of student projects that used GenAI for a level design prototype.



Fig. 3. A student first created a placeholder layout with defined interaction zones and navigation logic. After generating the background image with GenAI, the student revised the clickable regions to align with the final scene.



Fig. 4. A student with a Digital Media Technology background used AI to generate a cohesive set of 2D sprite assets, applying their visual judgment to guide asset selection and integration.

2) *Using AI to Extend Skills*: 3 out of 30 students used GenAI to extend specific skills. For novice programmers, AI provided code scaffolding that helped them learn new technologies. For students with visual or design training but limited illustration skills, GenAI-generated assets helped translate their aesthetic judgment into usable game materials. Fig. 4 demonstrates a student using GenAI to create art assets and learn about sprite sheets. In this case, GenAI served as a skill-extending tool, opening new pathways for learning and game production.

V. DISCUSSION

In this discussion, we draw on self-determination theory (SDT) to interpret the implications of our case study and thematic analysis of students’ learning in a GenAI-integrated game design and development classroom [11]. SDT explains how learning, engagement, and well-being are shaped by the fulfillment of three psychological needs. Autonomy refers to learners’ sense of ownership over their actions, competence refers to their sense of effectiveness and capability, and relatedness refers to their sense of connection with others. SDT provides a useful lens for examining how GenAI may support or undermine students’ design choices, skill development, collaboration, feedback, and classroom interaction.

Synthesizing the qualitative findings using SDT, we developed a heuristic framework to support agency and creativity with GenAI (see Table I). The framework identifies patterns of GenAI use that can either constrain or support students’ autonomy, competence, and relatedness in game design and development.

A. GenAI, Autonomy, and Competence

GenAI can support competence by helping students create playable prototypes, generate assets, and implement features that may otherwise be beyond their current technical or artistic skills. This helps novice developers to bring abstract concepts to functional game artifacts for playtesting and feedback. However, over-reliance on GenAI hurts autonomy. When students accepted AI outputs without evaluation or adjusted their design goals primarily based on what the tool could generate, the AI system became a stronger driver of the project’s direction. This suggests that, in addition to evaluating the final game product, educators should assess the development process to determine whether students can articulate their design decisions, justify their use of GenAI, and demonstrate an iterative workflow aligned with their creative goals.

B. Relatedness and Human Support

Despite GenAI’s immediate technical support and creative assistance, students still depended on peers and instructors to interpret, validate, and refine AI outputs—for example, debugging generated code or assessing whether generated assets fit the intended game experience. Human collaboration thus acts as a necessary complement to AI assistance, underscoring the role of relatedness in GenAI education. Learning with GenAI should not be framed as an isolated interaction between an individual student and an AI tool. Embedding AI work within collaborative learning structures such as peer review, team discussion, instructor critique, and shared reflection on AI use helps students contextualize AI outputs, retain authorship, and make more deliberate design choices.

C. Pedagogical Implications

We suggest that instructors introduce students to the common uses and misuses of GenAI early. Discussing these examples early can help students make more deliberate choices about when to use AI, when to revise its outputs, and when to

TABLE I
A HEURISTIC FRAMEWORK FOR SUPPORTING AGENCY AND CREATIVITY WITH GENAI

Dimension	Less Supportive Pattern	More Supportive Pattern	SDT Need(s)
1. Vision & Intentionality	Follow what GenAI can easily produce rather than the overarching design goal.	Define a design goal that guides how GenAI is used, revised, or rejected.	Autonomy
2. Evaluation & Curation	Use GenAI outputs with little review, leaving bugs, inconsistencies, or unclear logic.	Evaluate, select, modify, or reject GenAI outputs based on technical, aesthetic, and gameplay needs.	Competence
3. Problem Solving & Adaptation	Change or discard the design because GenAI cannot produce the desired result.	Revise prompts or seek alternatives to deliver the initial design goals.	Competence
4. Synthesis & Integration	Integrate GenAI content that is disconnected from the intended player experience.	Orchestrate GenAI and human-authored components into a coherent game experience.	Competence
5. Collaborative Reflection	Work solely with GenAI, with limited discussion of its use and limitations.	Discuss GenAI outputs, and ethical and design issues around GenAI and games.	Relatedness

rely on human expertise. The assessment of the class should emphasize both process and product. A polished game does not necessarily show that students understood or directed the creative process. Instead, students learn by reflecting on their experiences [12], as presented in lessons learned from professional training programs for game developers [13]. Reflection reports, design rationales, and documentation of AI workflows can provide insight into students' agency, creativity, and learning across the development process.

VI. FUTURE WORK & CONCLUSION

As an exploratory case study, this work has identified emerging patterns of work and learning in the adoption of GenAI in a game design and development class. Because this study focused on a single case and a limited number of participants, the findings should be interpreted as preliminary rather than generalizable. Future work is needed to deepen the understanding of GenAI's impact on skill development.

The findings suggest that learning to use GenAI in game development is not only a technical challenge, but also a matter of fostering developers' agency and creativity. By framing GenAI use through autonomy, competence, and relatedness, this paper offers an initial framework for helping students work with AI while maintaining creative control, authorship, and design judgment.

VII. ACKNOWLEDGEMENT

This study analyzes a course in which students were asked to use GenAI to produce assets, code, and text; AI-generated outputs are clearly labeled where cited or discussed. The authors used ChatGPT and DeepSeek only for translation, grammar checking, and copy editing, and all final content was reviewed and approved by the authors.

REFERENCES

- [1] G. Smith, J. Whitehead, and M. Mateas, "Tanagra: a mixed-initiative level design tool," in *Proceedings of the Fifth International Conference on the Foundations of Digital Games*, ser. FDG '10. New York, NY, USA: Association for Computing Machinery, Jun. 2010, pp. 209–216. [Online]. Available: <https://dl.acm.org/doi/10.1145/1822348.1822376>
- [2] A. Liapis, G. N. Yannakakis, and J. Togelius, "Sentient Sketchbook: Computer-aided game level authoring," May 2013.
- [3] J. Zhu, A. Liapis, S. Risi, R. Bidarra, and G. M. Youngblood, "Explainable AI for Designers: A Human-Centered Perspective on Mixed-Initiative Co-Creation," in *2018 IEEE Conference on Computational Intelligence and Games (CIG)*. Maastricht, Netherlands: IEEE Press, Aug. 2018, pp. 1–8. [Online]. Available: <https://doi.org/10.1109/CIG.2018.8490433>
- [4] D. Yang, E. Kleinman, and C. Hartevel, "GPT for Games: An Updated Scoping Review (2020-2024)," *IEEE Transactions on Games*, pp. 1–16, 2025. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/10974629>
- [5] R. Panchanadikar and G. Freeman, "I'm a Solo Developer but AI is My New Ill-Informed Co-Worker": Envisioning and Designing Generative AI to Support Indie Game Development," *Proc. ACM Hum.-Comput. Interact.*, vol. 8, no. CHI PLAY, pp. 317:1–317:26, Oct. 2024. [Online]. Available: <https://dl.acm.org/doi/10.1145/3677082>
- [6] —, "A Potential Teammate?: Understanding How Indie Game Developers Approach Generative AI's Involvement in Their Small-Scale Creative Teamwork," in *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, ser. CHI '26. New York, NY, USA: Association for Computing Machinery, Apr. 2026, pp. 1–15. [Online]. Available: <https://dl.acm.org/doi/10.1145/3772318.3790869>
- [7] J. D. Boucher, G. Smith, and Y. D. Tellier, "Is Resistance Futile?: Early Career Game Developers, Generative AI, and Ethical Skepticism," in *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, ser. CHI '24. New York, NY, USA: Association for Computing Machinery, May 2024, pp. 1–13. [Online]. Available: <https://dl.acm.org/doi/10.1145/3613904.3641889>
- [8] J. Boucher, M. Chen, G. Smith, and Y. D. Tellier, "Examining the Trajectory of Early Professionals' use of Generative AI in the Game Development Process from 2023 to 2024," Nov. 2024. [Online]. Available: <https://par.nsf.gov/biblio/10617574-examining-trajectory-early-professionals-use-generative-ai-game-development-process-from>
- [9] R. K. Yin, *Case Study Research: Design and Methods*. SAGE, 2009.
- [10] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative Research in Psychology*, vol. 3, no. 2, pp. 77–101, Jan. 2006. [Online]. Available: <https://www.tandfonline.com/doi/abs/10.1191/1478088706QP0630A>
- [11] R. M. Ryan and E. L. Deci, "Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being," *American Psychologist*, vol. 55, no. 1, pp. 68–78, 2000.
- [12] J. Dewey, *How We Think*. D.C. Heath & Company, 1910, google-Books-ID: WF0AAAAAMAAJ.
- [13] M. Chen and G. Smith, "Game Development as Project-Based Learning: Synthesizing Postmortems of Student-Created Mobile Games," in *Proceedings of the 19th International Conference on the Foundations of Digital Games*, ser. FDG '24. New York, NY, USA: Association for Computing Machinery, Jul. 2024, pp. 1–11. [Online]. Available: <https://dl.acm.org/doi/10.1145/3649921.3649999>